HarfBuzz Direct Rendering vs PDFKit

Method 1: PDFKit doc.text() (automatic)

אֱלֹהִים בָּרָא בְּרֵאשִׁית

Method 2: HarfBuzz direct glyph positioning

בְּרֵאשִׁית בָּרָא אֱלֹהִים

Analysis:

• Method 1 uses PDFKit's built-in text rendering (calls HarfBuzz internally)
• Method 2 demonstrates manual glyph positioning from HarfBuzz data
• Nikud marks should appear in both methods if font has GPOS tables
• For production: need PDFKit patch or direct PDF operators for glyphs

HarfBuzz Data:

• Total glyphs: 26
• Text length: 30 characters
• Font: Ezra SIL (2048 units/em)
• Scale: 0.007813
• Has X offsets: true (horizontal positioning)
• Has Y offsets: false (vertical positioning)
• Has zero-advance glyphs: true (marks)

Ezra SIL GPOS behavior:
• Nikud marks rendered as separate glyphs with xAdvance=0
• Horizontal position adjusted via xOffset (not yOffset)
• Vertical position assumed by renderer from glyph metrics
• This is valid GPOS behavior - positioning data is present!

Note: PDFKit v0.15.0 does not expose direct glyph drawing API. Production implementation requires:

1. Fork PDFKit with _writeGlyph() exposed, or
2. Write PDF operators directly (more control)
3. Use different PDF library (pdf-lib, pdfmake)